5G Expanding brackets

Learning Objective	Apply algebraic properties to rearrange, expand and factorise linear expressions
Today We Are	Expanding brackets
Indicators of Progress	• I can expand brackets

On your mini whiteboards:

1. Write the expanded form for each expression
2. Simplify the expression

(1) $2x^2 \times 4cx^2$

$= 2 \times 4 \times c \times x \times x \times x \times x$

$= 8cx^4$

(2) $\frac{6y^3}{12y^2}$

$= \frac{\cancel{6}^{1} \times y \times \cancel{y} \times \cancel{y}}{\cancel{12}^{2} \times \cancel{y} \times \cancel{y}}$

$= \frac{1 \times y}{2}$

$= \frac{y}{2}$

Remember:

- **Multiplying terms**: Any terms can be multiplied
- **Adding/ Subtracting terms**: Can only be done with like terms
- A x A = **A²**

What is the Distributive Law?

Roleplay Activity: Dinner Party

- **3** Volunteers (1 guest, 2 dinner hosts)
- **Step 1:** The guest must **knock** on the hosts door.
- **Step 2:** The guest must **greet** the hosts with a fist bump or handshake

$= 3 \times x + 3 \times 1$

$= 3x + 3$

$= 3 \times 2x + 3 \times 5$

$= 6x + 15$

$= 4x \times 2 - 4x \times y$

$= 8x - 4xy$

$= -8 \times 7 + -8 \times 2y$

$= -56 - 16y$

$= 12xy + 7x \times 2 + 7x \times -y$

$= 12xy + 14x - 7xy$

$= 5xy + 14x$